

QA DISPOSITION REPORT burn-in parametric screening

Component	LOT203-C0246	Lot	LOT203	Board	LOT203-B03
Socket	r6 c6	Model version	dataset-v1.1 / ModuleA-d13e1ef / ModuleB-2628c092619:01		

DECISION: REVIEW

LAYER 1 — deterministic evidence (every value cites its comparison basis)

Absolute limit	PASS (79.54 nA vs 100.0 nA max, leakage_na)
Lot-relative level	+8.9 robust sigma vs lot median on leakage_na (median 27.26 nA, sigma 5.89 nA, n=497, MAD estimator; AEC-Q001 IQR/1.35 gives 5.81 nA)
Drift 0h to 24h	+18.364 nA (+90.4%) on leakage_na; lot median drift +1.281 nA (+5.5%), part is +14.4 robust sigma on the lot's drift distribution
Drift rate	0.7652 nA/h (lot median 0.0534 nA/h, 14.3x)
Max interval jump	+29.4 robust sigma on leakage_na; largest interval change is 1.6x the median of the others (spread -> progressive drift)
Predicted 168h	45.443 nA (leakage_na, from 0h and 24h only)
95% upper bound	76.075 nA -> EXCEEDS lot-derived safe limit 64.182 nA
Joint anomaly	Mahalanobis D2 = 224.6 (good-part 99th pct = 14.5); largest contribution leakage_na at 81% of the total
Leakage/delay sign	leakage +8.9 sigma, delay -0.2 sigma -- consistent with the lot's process locus
Spatial check	no neighbour clustering on board LOT203-B03 (3/64 flagged), component-specific
Data quality	all checkpoints valid
Primary reason	abnormal drift rate relative to lot peers
Secondary	elevated level relative to lot peers
Action	Hold. Repeat the measurement to rule out test error, then refer to a reliability engineer with this report.

LAYER 2 — model attribution

detector	normalised strength	fired	feature (Module B)	SHAP (nA)
L3b_T2	71.62	YES	v24	+8.795
L4d_AutoEnc	66.69	YES	d1	+3.967
L3c_kNN	23.39	YES	r1	+1.945
L3b_Q	22.19	YES	z:d1	+0.425
L4b_LOF	18.29	YES	v0	-0.187

Detector strength is distance past that detector's own operating threshold (1.0 = exactly at it). SHAP values are exact TreeSHAP from the LightGBM drift model, in target units. **Caveat:** v0 and v24 are 0.99 correlated, so SHAP credit between them is not identifiable — a Huber model of equal accuracy attributes it the other way round. Treat Layer 2 as supporting evidence; Layer 1 is the auditable record.

COUNTERFACTUAL

No single-parameter change clears the rules; the evidence is distributed across several measurements.

LAYER 3 — failure-mechanism HYPOTHESIS (for the FA engineer; not a diagnosis)

No mechanism signature matched; no hypothesis offered.

SIGN-OFF

Inspector		Date	
Disposition		Signature	
Notes			

Decision support only. Final rejection remains under approved engineering rules and QA control; this system detects, ranks, explains and forecasts. The model never overrides a hard engineering limit in either direction.